Emediong Akpan

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Profile Summary

Physics Expert (Ph.D., MIT) specializing in Theoretical and Computational Physics with strong experience in quantum mechanics, data modeling, and scientific computing. Proven ability to develop large-scale physical simulations, author peer-reviewed research, and contribute subject-matter expertise to AI and machine learning projects. Passionate about advancing science-driven innovation and computational solutions.

Education

Massachusetts Institute of Technology (MIT) – Cambridge, MA

Ph.D. in Physics (Theoretical & Computational Physics), 2018–2024
Master of Science in Physics, 2016–2018
Research focus: Quantum mechanics, computational modeling, and advanced problem-solving.

Work Experience

Graduate Research Assistant – MIT, Physics Department (2018–2024)
Conducted original research in computational physics and large-scale modeling. Published findings in peer-reviewed journals; contributed to model development for AI integration. Collaborated with interdisciplinary teams to improve computational frameworks.

Teaching Assistant – MIT (2019–2023)
Designed and graded problem sets in advanced physics subjects. Mentored students and provided academic support in research methodology. Recognized for clarity and teaching excellence.

Research Consultant – Physics Modeling (Freelance / Remote, 2023–Present)
Provided domain expertise for AI/ML-based physics simulations. Built physics-specific datasets for machine learning applications. Authored technical documentation and research formulations.

Projects

AI Interviewer (2024)
Developed a chatbot to assist users with structured interview preparation using NLP and simulation logic.

Publications

Quantum State Modeling with Large-Scale Simulation – Journal of Theoretical Physics (July 13, 2024)

Novel Computational Methods in Thermodynamic Systems – Applied Physics Letters (July 13, 2024)

Awards

• Recipient of Graduate Fellowship / Scholarship for Research Excellence – MIT
• Teaching Assistant of the Year – MIT Physics Department
• Best Paper Award – Journal of Theoretical Physics

Certifications

• Machine Learning – Stanford University (Coursera)
• Deep Learning Specialization – DeepLearning.AI
• Python for Data Science – IBM / edX
• MATLAB for Engineers – MathWorks
• APS & IOP Professional Memberships

Skills

Core Physics: Quantum Mechanics, Electromagnetism, Statistical Mechanics, Computational Physics
Technical Tools: Python, MATLAB, Mathematica, LaTeX, Simulation Software
Research & Writing: Academic Writing, Peer-Reviewed Publishing, Problem Curation
Soft Skills: Analytical Thinking, Communication, Attention to Detail, Remote Collaboration